

A Comparison of Supervised Learning Models on Distinct Datasets for Malware Detection

MALWARE, MEMORY ANALYSIS, ANDROID, MACHINE LEARNING

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OBJECTIVE

Evaluate how well supervised ML models perform on very different malware types:

Memory-resident (RAM-based)

Android malware

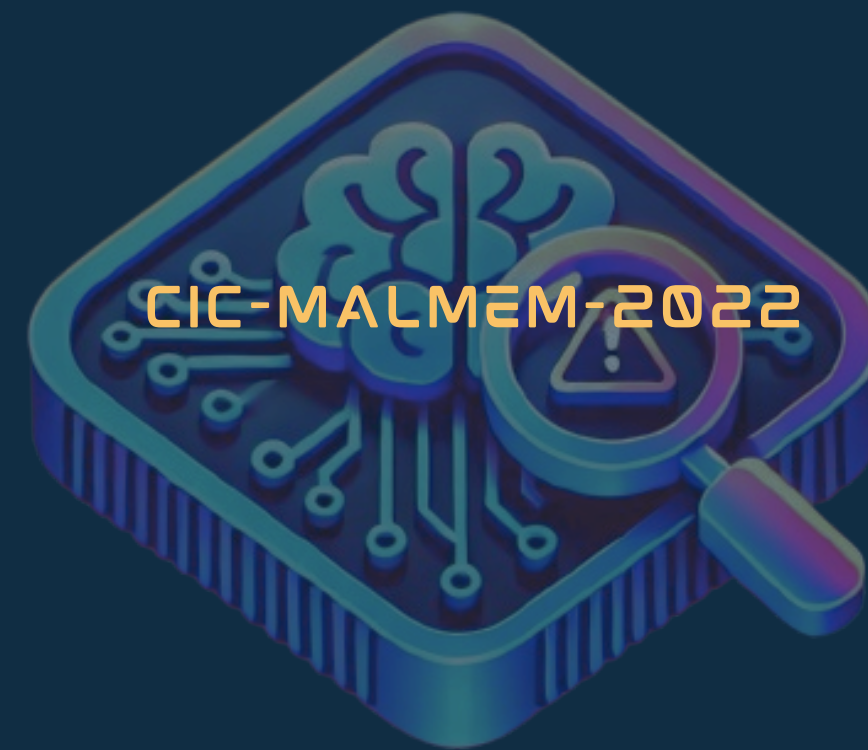
INTRODUCTION

Malware is more sophisticated than ever:
fileless and Android-based threats are rising

Signature-based detection is no longer enough

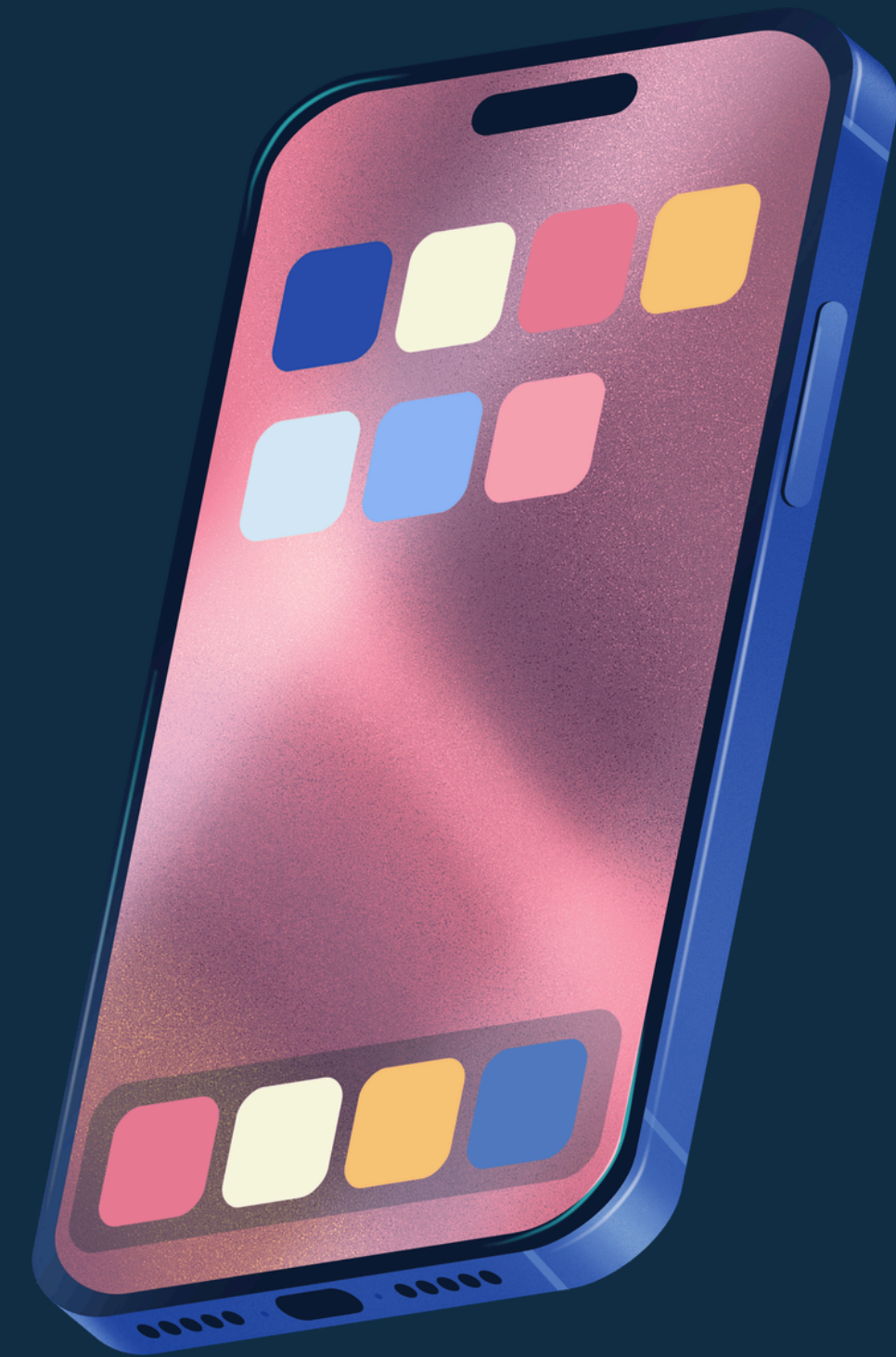


DATASETS



CIC-MALDROID-2020

INSTITUTION	Canadian Institute for Cybersecurity (CIC)
RELEASE YEAR	2020
TOTAL ANDROID APPS	17 341
MALICIOUS	11 860
BENIGN	5 481



DATASETS

CIC-MALDROID-2020



TYPES OF FEATURES INCLUDED

The dataset includes over 400 features, which come from two main analysis methods:

STATIC FEATURES

- Permissions requested
- API calls used
- App Components
- Intent Filters
- Strings and metadata embedded in the code
- Use of specific libraries or system functions

DYNAMIC FEATURES

- File system activity
- System logs and error messages
- Network activity
- Use of system resources
- Execution patterns

DATASETS

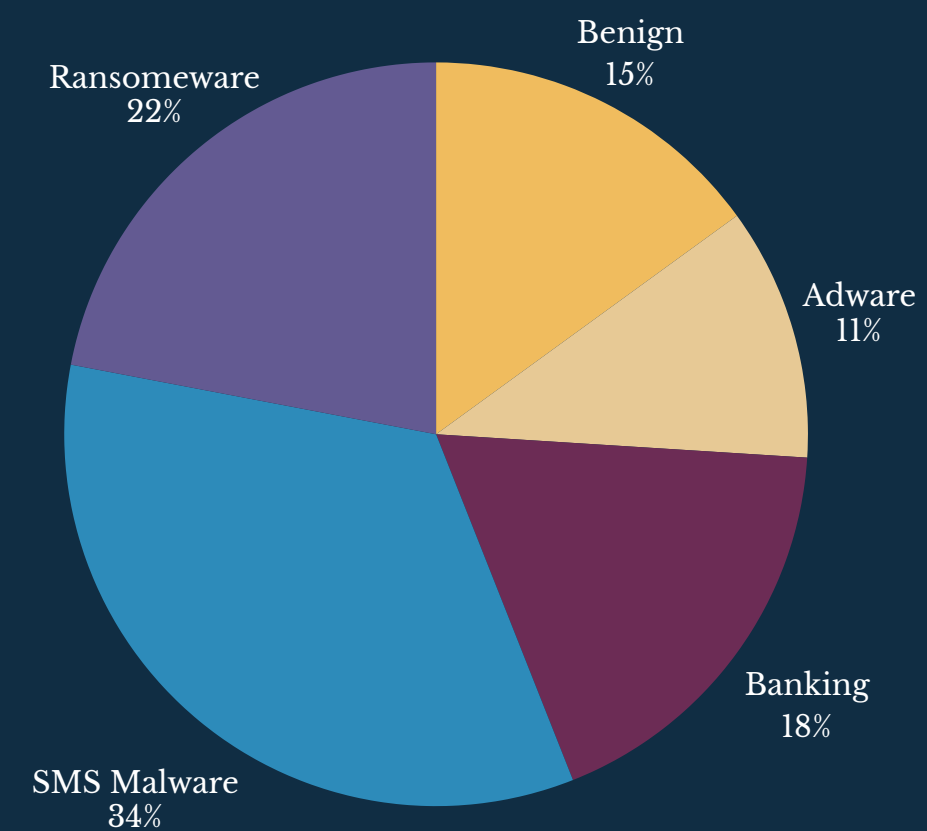
CIC-MALROID-2020



PRIMARY OBJECTIVE OF THE DATASET

Provide a comprehensive dataset for **Android malware detection**, enabling the development and evaluation of machine learning and deep learning techniques.

MALWARE CATEGORIES

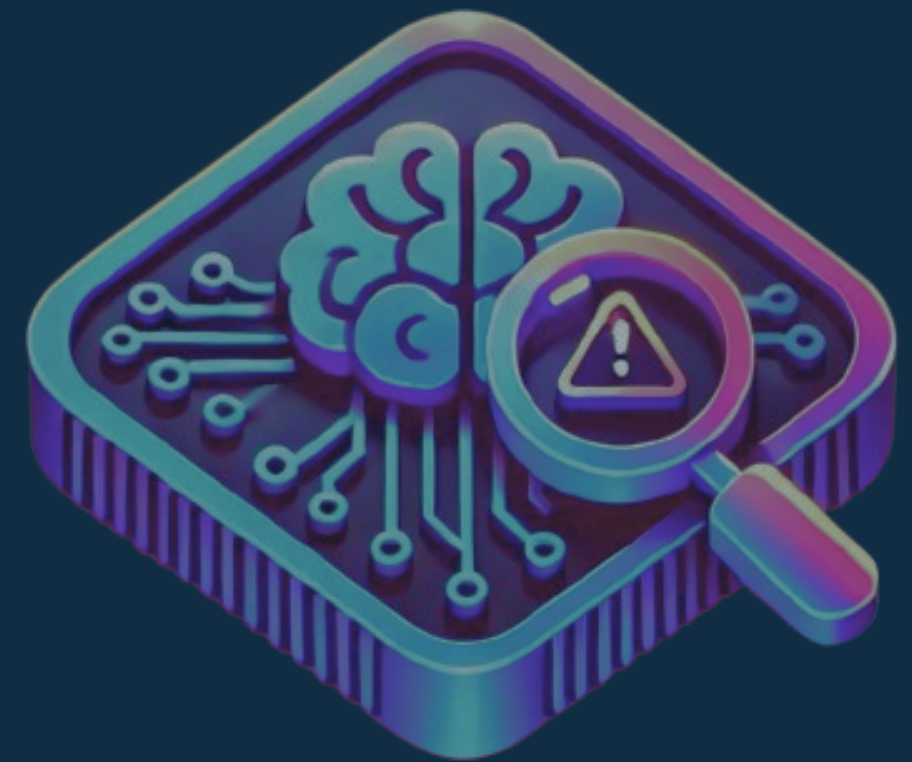


CIC-MALMEM-2022

INSTITUTION Canadian Institute for Cybersecurity (CIC)

RELEASE YEAR 2022

MEMORY FORENSIC DATA ~20,000+ process samples





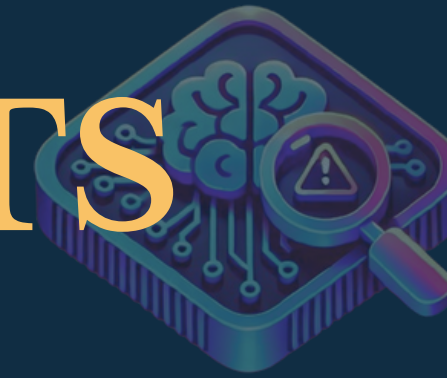
TYPES OF FEATURES INCLUDED

Each row in the dataset represents a process running in memory and includes dynamic features extracted through memory forensics.

Process Metadata
API calls used
Memory Usage
Handles and Objects
DLLs and Modules
Threads
CPU/Runtime Behavior

DATASETS

CIC-MALMEM-2022

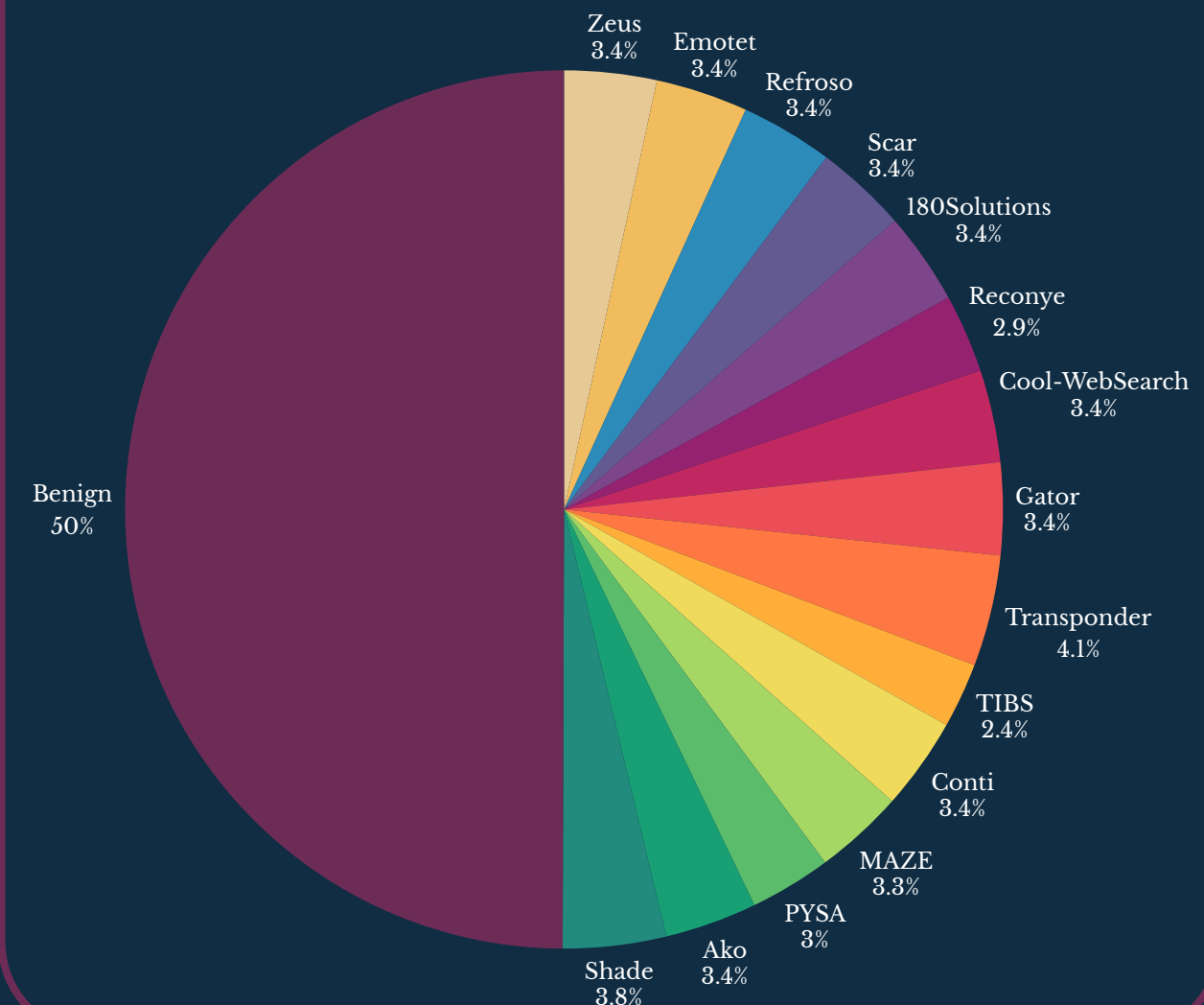


PRIMARY OBJECTIVE OF THE DATASET

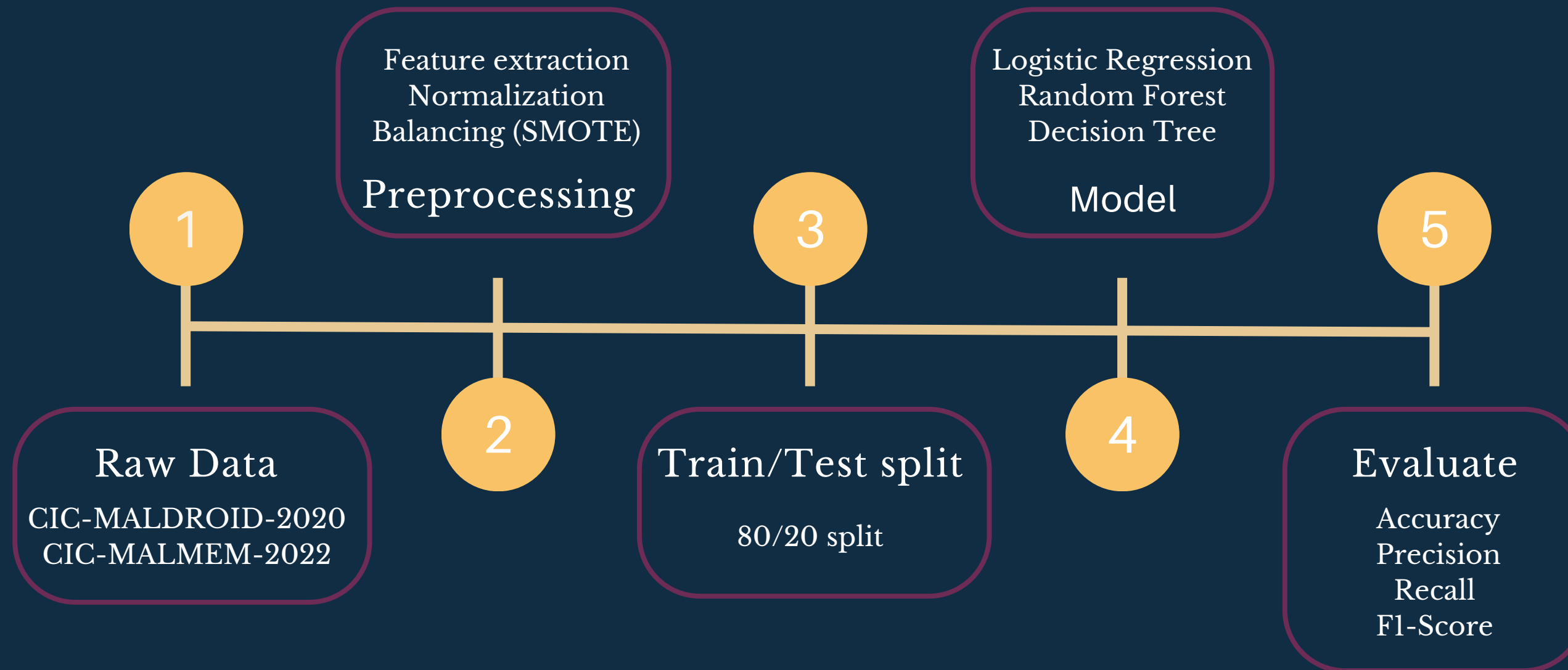
Enable robust and behavior-based detection of Windows malware using **memory forensics**.

Focuses on detecting active malware processes in system memory (RAM).

MALWARE CATEGORIES



OUR PIPELINE



BINARY CLASSIFICATION RESULTS

DEFAULT MODEL HYPERPARAMETERS

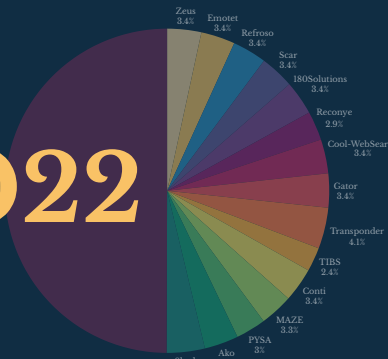
CIC-MALMEM-2022

MODEL	ACCURACY	PRECISION	RECALL	F1-SCORE
LOGISTIC REGRESSION	0.9988	0.9988	0.9988	0.9988
RANDOM FOREST	0.9999	0.9999	0.9999	0.9999
DECISION TREE	0.9999	0.9999	0.9999	0.9999

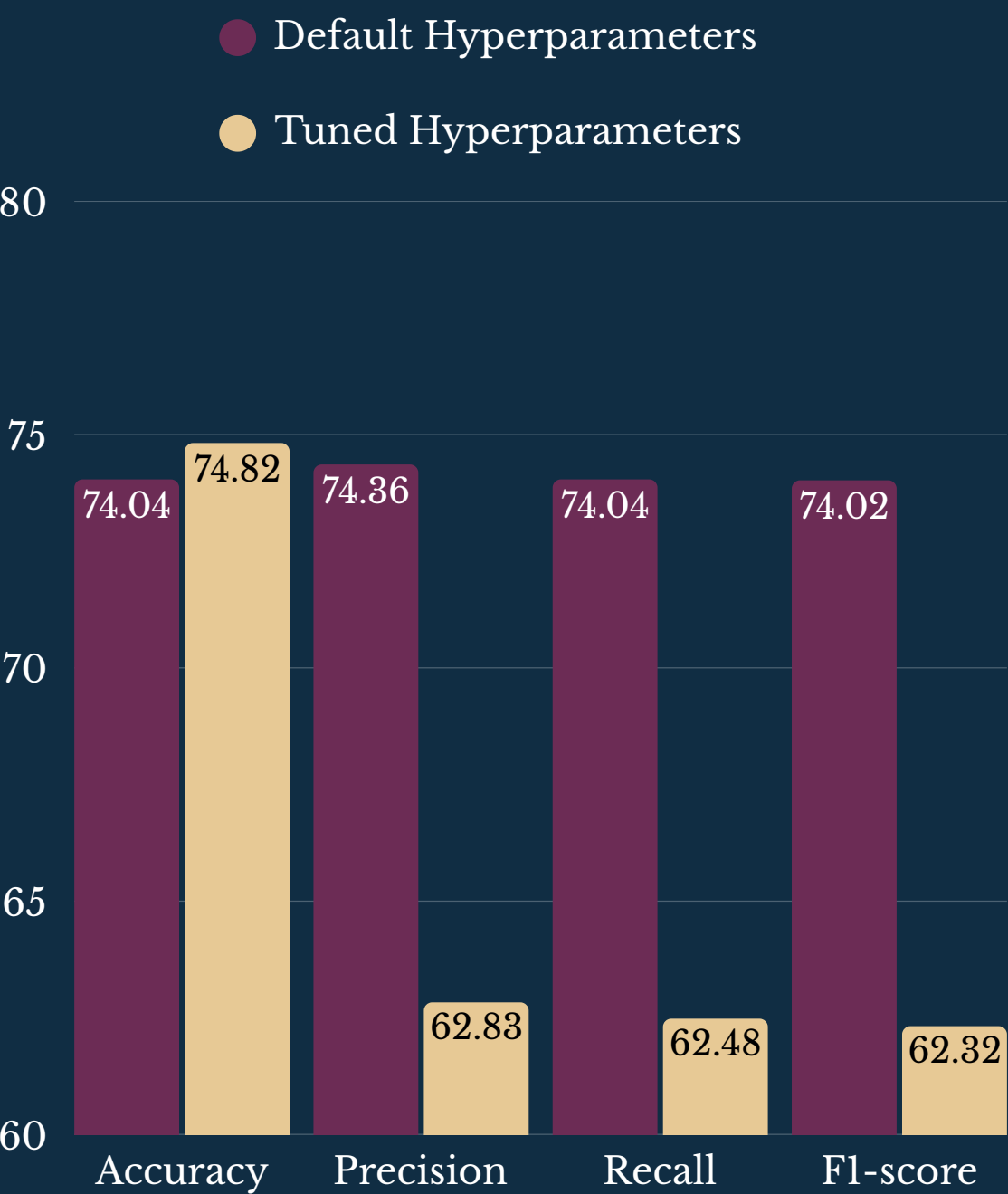
CIC-MALDROID-2020

ACCURACY	PRECISION	RECALL	F1-SCORE
0.7551	0.7602	0.7551	0.7459
0.9340	0.9350	0.9340	0.9340
0.8926	0.8925	0.8926	0.8925

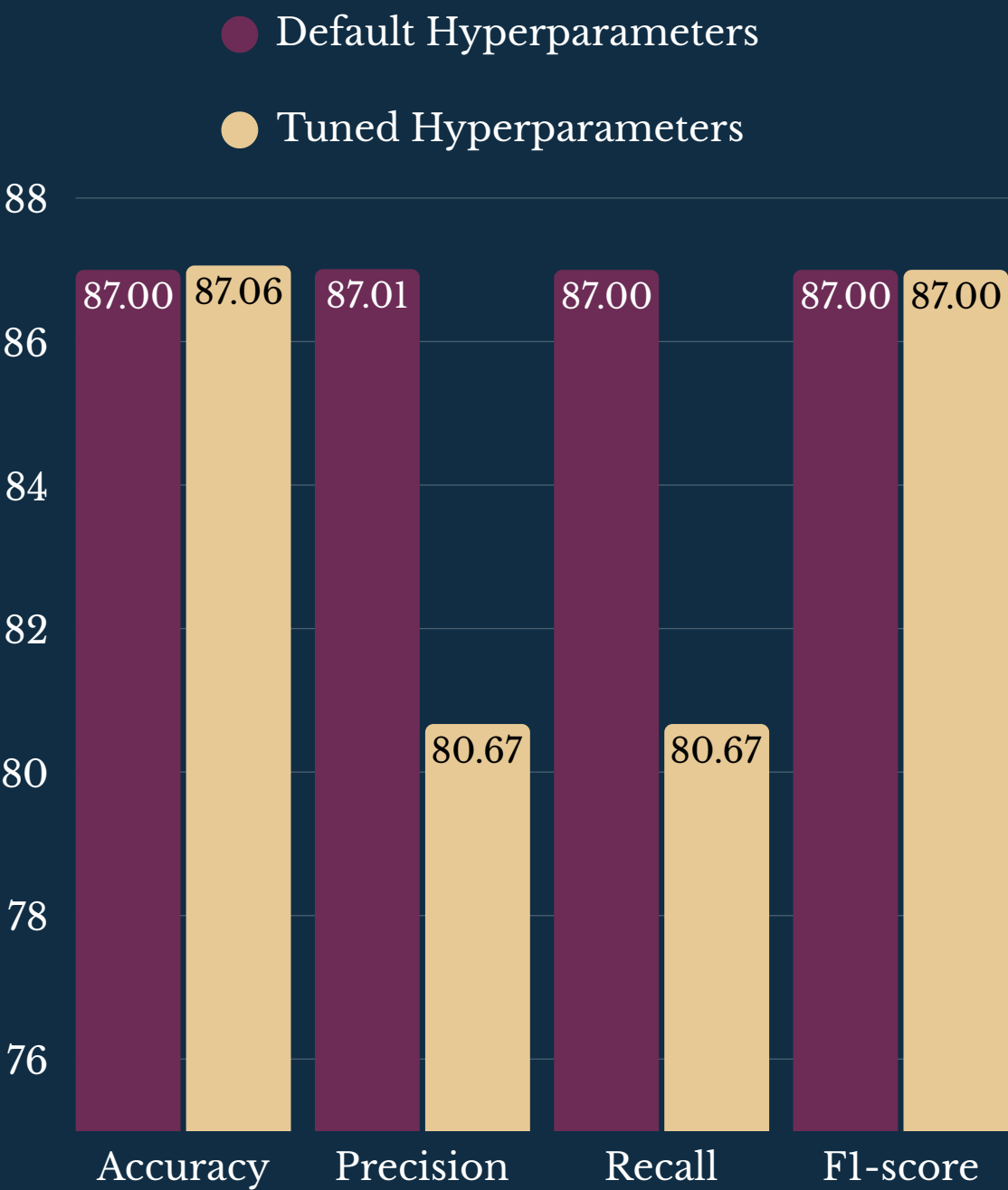
SUB FAMILY CLASSIFICATION RESULTS CIC-MALMEM-2022



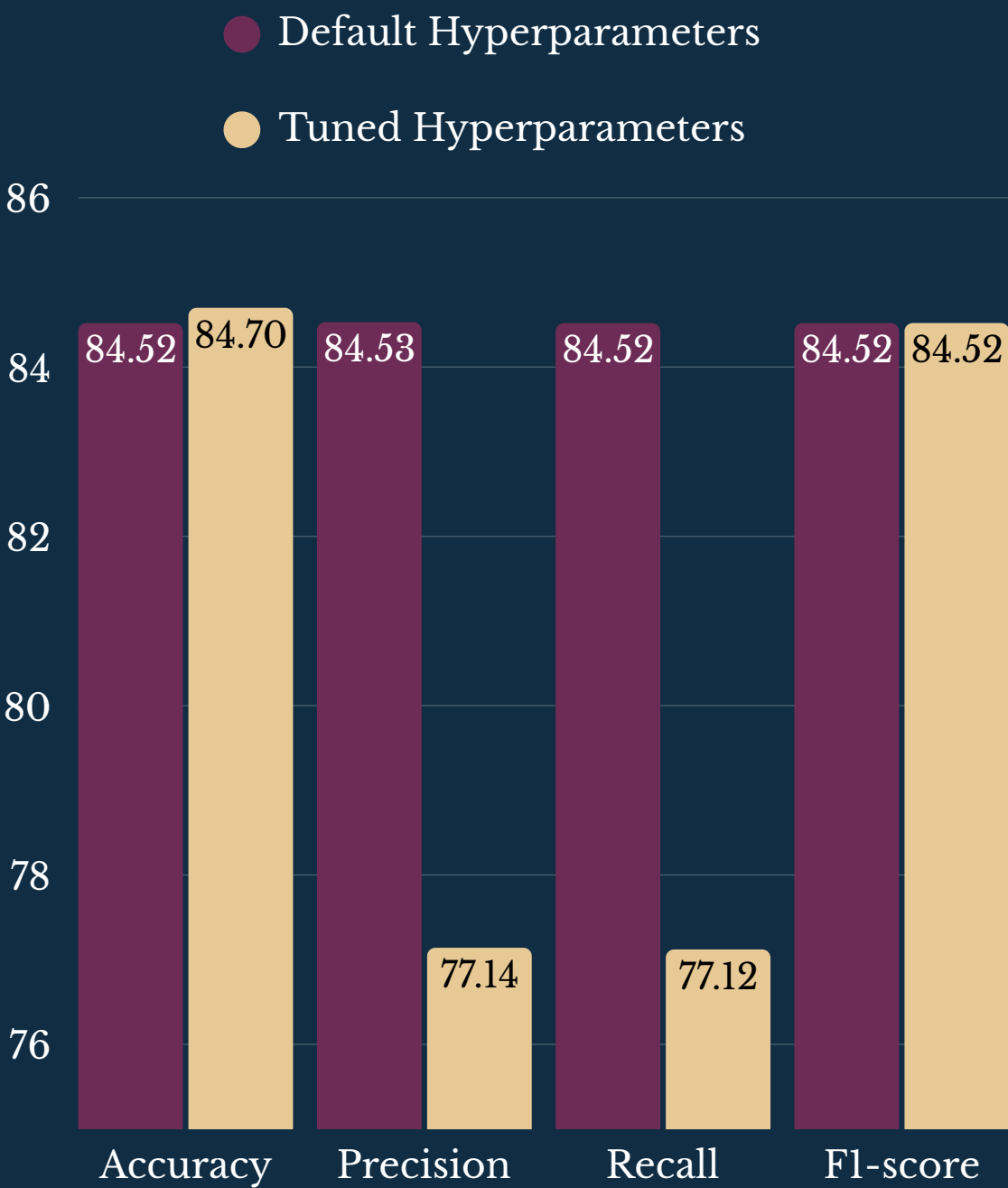
LOGISTIC REGRESSION



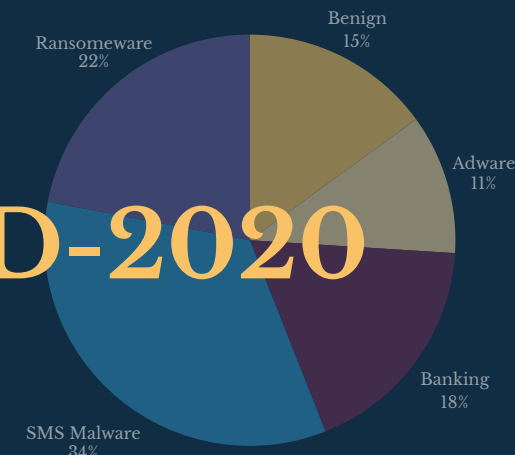
RANDOM FOREST



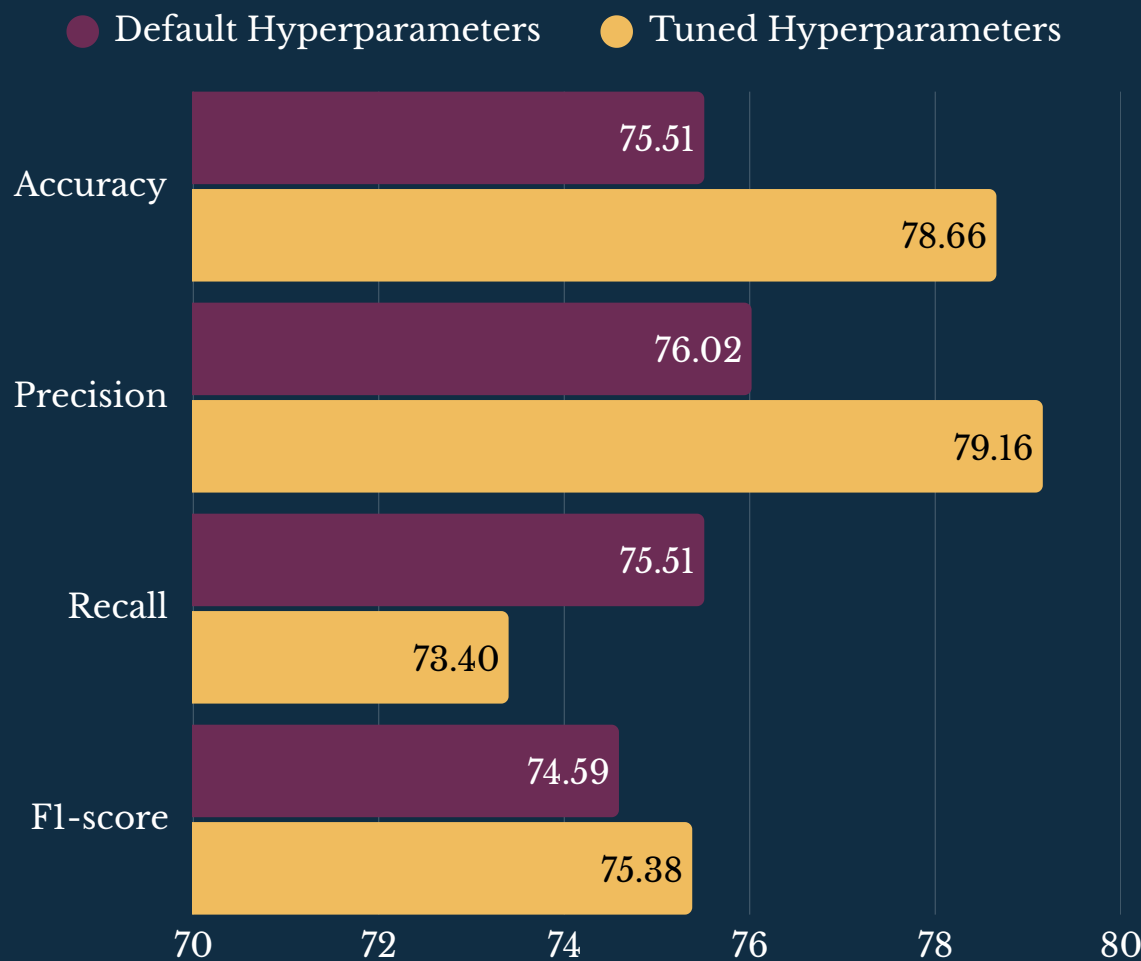
DECISION TREE



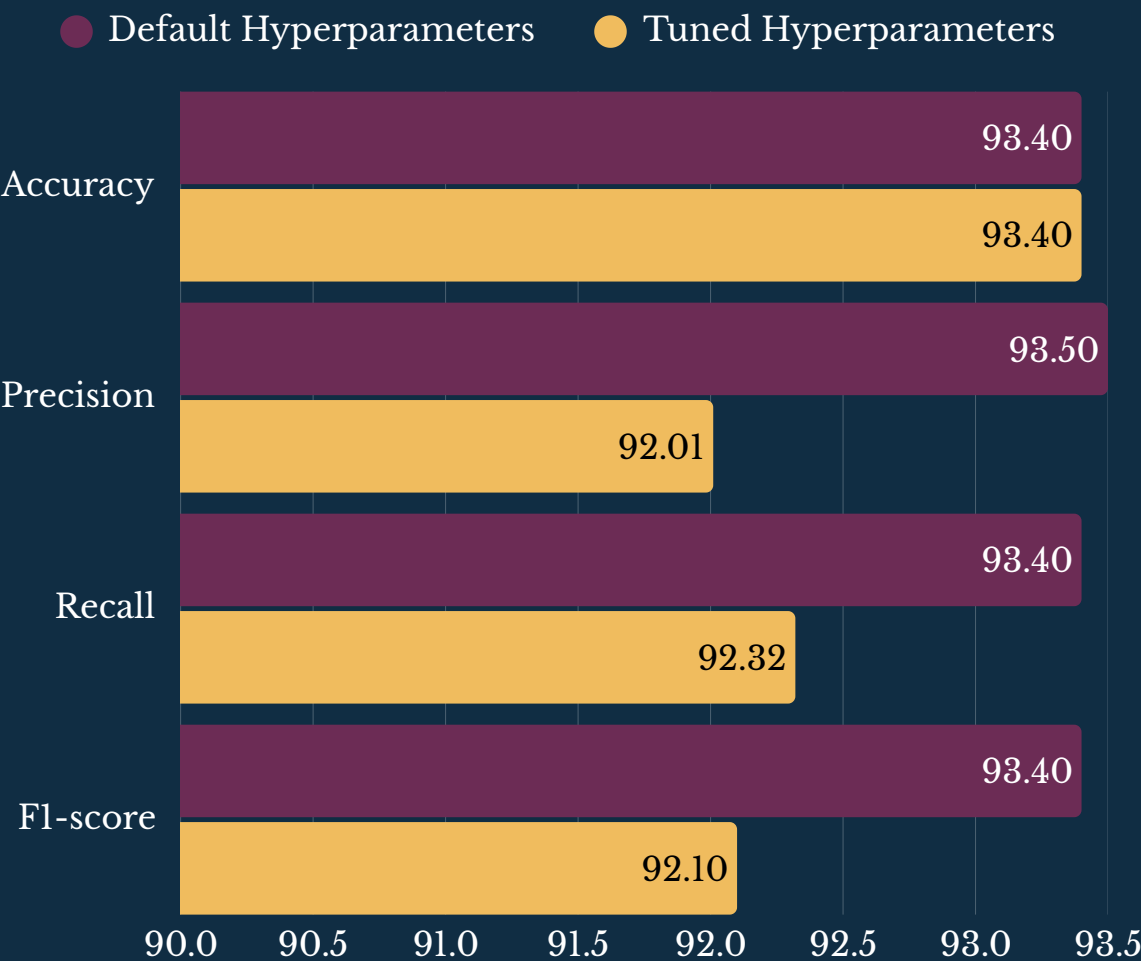
SUB FAMILY CLASSIFICATION RESULTS CIC-MALDROID-2020



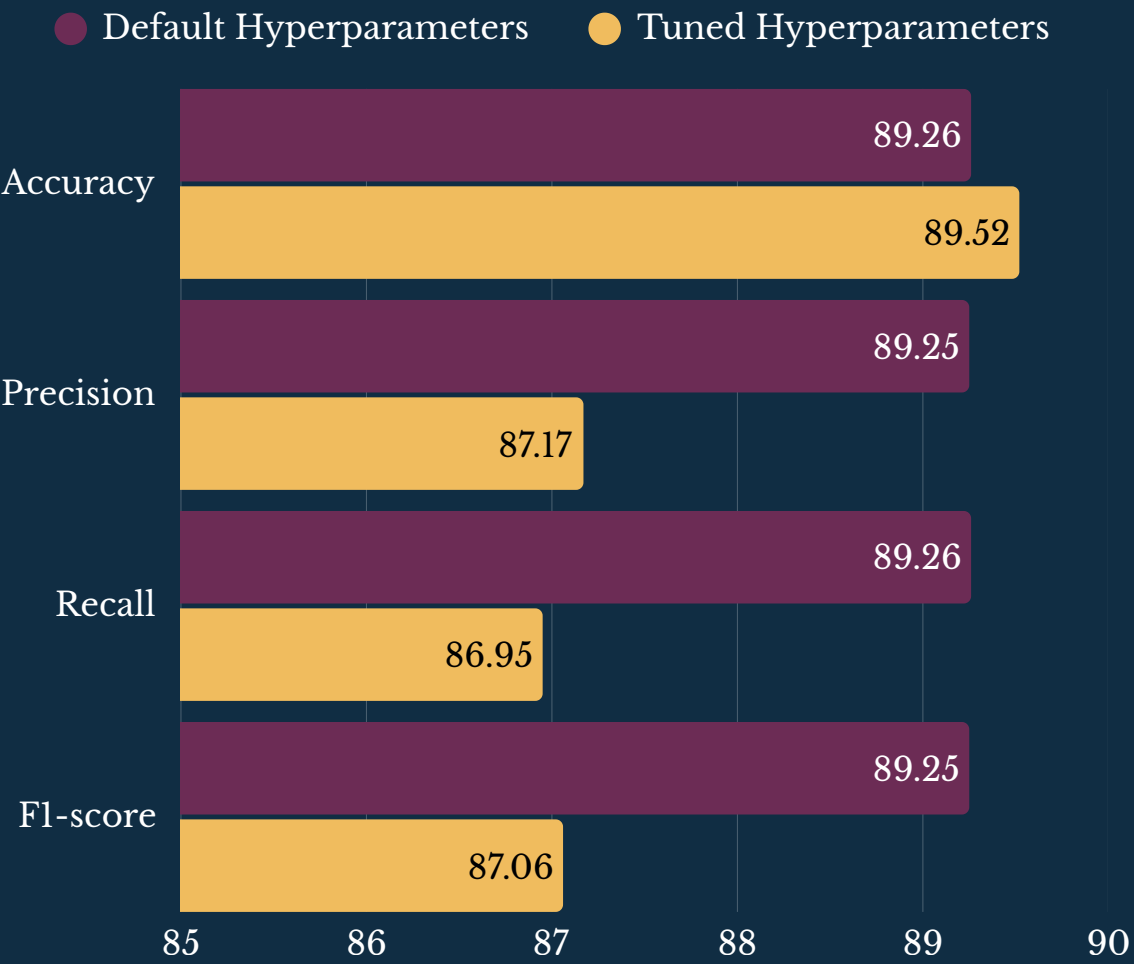
LOGISTIC REGRESSION



RANDOM FOREST



DECISION TREE



HYPERPARAMETER TUNING GAINS

LOGISTIC REGRESSION Minor impact, especially on Android dataset

RANDOM FOREST Some precision/recall gains

DECISION TREE Better on memory dataset after tuning



CONCLUSION AND FUTURE WORK



Random Forest is the most reliable across tasks

Dataset matters: structured vs unstructured affects results

Logistic Regression is a good baseline, ideal for fast inference

Explore deep learning models

Try dimensionality reduction & feature selection

Move toward real-time malware detection systems

A COMPARISON OF SUPERVISED LEARNING MODELS ON DISTINCT DATASETS FOR MALWARE DETECTION

THANK YOU !



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